Jonnalagadda Naveen Reddy

Software Engineer

9121905200 | naveenreddy0911@gmail.com | linkedin.com/in/naveenreddy0911 | github.com/naveenreddy0911

Professional Summary

IIT Kharagpur graduate (M.Tech & B.Tech) experienced in designing and building software services at scale using C++, Python, JavaScript, and TypeScript. Strong in object-oriented design, system and integration testing, and translating stakeholder requirements into reliable, production-ready features. Background in performance optimization, cross-functional delivery, and using AI-powered tools to improve development efficiency.

Education

Indian Institute of Technology, Kharagpur **Jul 2024 – May 2025**
Master of Technology (M.Tech) in Mechanical Engineering, CGPA: 8.19 *West Bengal, India*

Indian Institute of Technology, Kharagpur **Dec 2020 – May 2024**
Bachelor of Technology (B.Tech) in Mechanical Engineering, CGPA: 8.19 *West Bengal, India*

Work Experience

AI Engineer **Nov 2025 – Present**
Ascendion *Hyderabad, India*

- Design and develop **Python** backend services and **REST APIs (FastAPI)** with testable architecture, processing high-volume data from **5+** concurrent sources while maintaining data controls and reliability.
- Built automated pipelines and reporting services at scale, reducing manual analysis time by over **90%** across **4+** recurring workflows through efficient, reusable backend design.
- Built a web-based validation platform (**DeepEval**) used by model reviewers and governance teams to independently assess conversational and **RAG** chatbot quality across **100+** evaluation scenarios and metrics.
- Partnered with **3+** cross-functional stakeholder teams to gather requirements, deliver features, and improve reporting efficiency using modern engineering practices and AI-assisted development workflows.

Software Engineer Intern **Jun 2025 – Sep 2025**
Navea Ventures *Hyderabad, India*

- Designed and implemented backend services with **FastAPI** and **PostgreSQL**, applying object-oriented design and software architecture to model, store, and serve data across **10+** multi-entity domains.
- Translated business requirements into production features, delivering a **React/TypeScript** UI with **30+** pages integrated with **50+** backend service routes.
- Contributed to platform feature development by breaking down requirements, reusing shared components, and shipping tested code with **JSON schema validation** and **OAuth 2.0** security controls.
- Collaborated with cross-functional teams on system and integration testing, rapid prototyping, and end-to-end delivery of scalable application features.

Technical Skills

Languages: C++, Python, JavaScript, TypeScript, SQL

Software Engineering: Object-Oriented Design, Data Structures & Algorithms, System & Integration Testing, Performance Optimization

Development: Node.js, Express.js, FastAPI, REST APIs, React, HTML, CSS

Databases: PostgreSQL, MySQL, MongoDB, Data Modeling, Query Optimization

Tools: Git, GitHub, Azure DevOps, Docker, Linux, VS Code

Projects

Yapper | Software Services & REST APIs

- Designed and implemented **12+ REST APIs** and backend services with **Node.js/Express** and **PostgreSQL**, using object-oriented patterns and a repository-layer architecture across **4** relational tables.
- Built production-ready user-facing features with input validation (**Zod**), secure auth (**JWT/OAuth 2.0**), and end-to-end testing across **8+** flows including signup, login, and session management.

Master's Thesis | Real-Time System & Performance Optimization

- Built an end-to-end real-time detection pipeline in **Python** with **YOLOv8**, **VGG16**, and **ViT**, achieving **99.84%** accuracy and **94.5% mAP@50** on a **10+** class dataset.
- Improved system efficiency through transfer learning and performance tuning, reducing training time by **63%** while maintaining reliable inference under real-time constraints.

Awards and Achievements

- Secured AIR **2481** in JEE Main, AIR **3091** in JEE Advanced, and a State Rank **381** in TS EAMCET.
- Earned **District 1st** place in the National Anveshika Experimental Skills Test (NAEST).